

Advanced (Habanero)

Q1. Solve the inequality $|x| < 5$.

(A) $(-5, 5)$

(B) $(-\infty, -5) \cup (5, \infty)$

(C) $[-5, 5]$

(D) $(-\infty, 5]$

(E) $[5, \infty)$

(D) $x > 3$

(E) $x < -3$

Q2. Solve the inequality $|2x - 3| \geq 4$.

(A) $(-\infty, -0.5] \cup [3.5, \infty)$

(B) $(-\infty, -3.5] \cup [0.5, \infty)$

(C) $[-3.5, 0.5]$

(D) $(-\infty, 0.5]$

(E) $[3.5, \infty)$

Q6. The Absolute Value of the difference between x and 3 is greater than or equal to 4.

(A) $x \leq -1$ or $x \geq 7$

(B) $-1 < x < 7$

(C) $-1 \leq x \leq 7$

(D) $x > 7$

(E) $x < -1$

Q3. Graph the solution set of $|x + 2| > 3$ on a number line.

(A) $(-\infty, -5)$

(B) $(-5, 1)$

(C) $(-\infty, -5) \cup (1, \infty)$

(D) $[-5, 1]$

(E) $(-5, \infty)$

(A) $-\frac{1}{3} \leq x \leq \frac{1}{3}$

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(A) $-\frac{1}{3} < x < \frac{1}{3}$

(A) $[-1, 1]$

Q4. Solve the inequality $|x^2 - 4| < 5$.

(A) $(-3, 3)$

(B) $(-\infty, -3) \cup (-1, 1) \cup (3, \infty)$

(C) $[-3, -1] \cup [1, 3]$

(D) $(-\infty, -3]$

(E) $[3, \infty)$

Q5. The Absolute Value of x is at most 3.

(A) $-3 \leq x < 3$

(B) $-3 \leq x \leq 3$

(C) $-3 < x \leq 3$

Q7. Solve the inequality $|3x + 1| \leq 2$.

(A) $-\frac{1}{3} \leq x \leq \frac{1}{3}$

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$$[-1, 1]$$

Q8. A company packages its product in boxes that have a length that does not differ from 36 by more than 2.

(A) $|x - 36| > 2$

(B) $|x - 36| < 2$

(C) $|x - 36| = 2$

(D) $x > 38$

(E) $x < 34$

Open Ended Questions (FRQ)

Q9. Solve the inequality $|x^2 - 9| \geq 4$ and express the solution in interval notation.Q10. Model the situation using an absolute value inequality: The temperature in a greenhouse must not vary by more than 5°C from 75°C .

Chef's Tips

- To ensure students answer all questions, consider providing a template or sample answers.
- Encourage students to show all their work for FRQs and provide clear explanations for MQs.
- Remind students that interval notation and number line graphs are valid solutions for absolute value inequalities.